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TASK 1: SINGLE LEVEL INHERITANCE CODE

Main.java	Run	Output
<pre>1- class Animal { 2- Animal() { 3- System.out.println("Animal created"); 4- } 5- 6- void eat() { 7- System.out.println("Animal is eating"); 8- } 9- 10- void sleep() { 11- System.out.println("Animal is sleeping"); 12- } 13- } 14 15- class Dog extends Animal { 16- Dog() { 17- System.out.println("Dog created"); 18- } 19- 20- void bark() { 21- System.out.println("Dog is barking"); 22- } 23- } 24 25- class Main { 26- public static void main(String[] args) { 27- Dog d = new Dog(); 28- 29- d.bark(); 30- d.eat(); 31- d.sleep(); 32- } 33- }</pre>		<pre>Animal created Dog created Dog is barking Animal is eating Animal is sleeping === Code Execution Successful ===</pre>

TASK 2: SIGLE LEVEL INHERITANCE OF UNIVERSITY MANAGEMENT

```
Main.java  [Icons]  [Share]  [Run]  Output

1- import java.util.Scanner;
2
3- class University {
4     Scanner s = new Scanner(System.in);
5     String a;
6     int b;
7     String c;
8
9- void dep() {
10     System.out.println("Enter Your Department:");
11     a = s.nextLine();
12 }
13
14- void sem() {
15     System.out.println("Enter Your Semester:");
16     b = s.nextInt();
17     s.nextLine(); // clear buffer (important fix)
18 }
19 }
20
21- class Faculty extends University {
22- void f_name() {
23     System.out.println("Enter Your Faculty Name:");
24     c = s.nextLine();
25 }
26 }
27
```

```
Enter Your Department:
Ai
Enter Your Semester:
2
Enter Your Faculty Name:
computing

----- Student Information -----
Department: Ai
Semester: 2
Faculty Name: computing

=== Code Execution Successful ===
```

SINGLE LEVEL INHERITANCE OF SHOP MANAGER:

```
1- import java.util.Scanner;
2
3- class Shop {
4     Scanner s = new Scanner(System.in);
5
6+    void items() {
7        System.out.println("----- Welcome To Store -----");
8        System.out.println("Items:");
9        System.out.println("1. Chocolates\n2. Biscuits\n3. Snacks\n4. Cookies");
10       System.out.println("-----");
11    }
12
13+    void prices() {
14        System.out.println("Prices:");
15        System.out.println("1. Chocolates - 50 PKR\n2. Biscuits - 100 PKR\n3. Snacks - 150
16           PKR\n4. Cookies - 20 PKR");
17        System.out.println("-----");
18    }
19
20- class Customer extends Shop {
21     String item = "Snacks";
22     int qty = 5;
```

----- Welcome To Store -----
Items:
1. Chocolates
2. Biscuits
3. Snacks
4. Cookies

Prices:
1. Chocolates - 50 PKR
2. Biscuits - 100 PKR
3. Snacks - 150 PKR
4. Cookies - 20 PKR

Customer Order Is: Snacks
Qty: 5
Bill: 750 PKR

=== Code Execution Successful ===